



Yatendra Kumar Singh | 23CL60R07

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EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2025	M Tech (CORAL)	IIT Kharagpur	8.16 / 10
2022	B Tech (Civil Engineering)	Anand Engineering College	8.01 / 10
2018	Intermediate	M D Jain Inter College	76.40%
2016	High School	St Andrews Public School	8.2 / 10

WORK EXPERIENCE

Post Graduate Engineer Trainee (PGET) | Larsen & Toubro | Environment, Health & Safety Engineer [June 2025- present]
•Currently employed in sustainability department | Reviewing TCFD, BRSR reports | Scenario Analysis | Net zero roadmap.
•Case study of Business Responsibility & Sustainability Report of Larsen & Toubro (2024-25) and Dabur India Limited (2023-24).

Project Engineer | Wipro [May 2022- July 2023]
•Designed a diabetes risk detection data from 441 individuals, targeting early identification of Type-2 diabetes based on behavioral and clinical risk factors.
•Evaluated six ML models (DT, LR, SVM, GB, RF, XGBoost) and deployed a custom ensemble model combining DT, RF, and XGBoost.
•Ensemble model achieved the highest accuracy of 91.3%, precision of 93.2%, and recall of 91.3%, age, extreme thirst, and family history as top predictors.

INTERNSHIPS

Summer Intern | Indian Institute of Tropical Meteorology (IITM) Pune | Dr. Bipin Kumar, Scientist 'F' [May'24 –Jul'24]
Project Title: *Bias Correction of Monthly Precipitation Estimates using Convolutional Neural Network 3D Deep Learning Model*
•Preprocessing involved standardizing grid values to a 32×32 format, resulting in generation of 1,290 input samples for each dataset spanning over 30 years.
•The Convolutional Neural Network (CNN) model was then compiled using the 'Adam' optimizer, MAE serving as the loss function and evaluation metric.
•This project enhanced my understanding of climate datasets, particularly through the preprocessing of .nc files for effective model input.

Machine Learning Internship (DS & AI) | YBI Foundation | Dr. Alok Yadav [Jul'24-Aug'24]
Project Title: *Big Sales Prediction using Random Forest Regressor Algorithm*
•The Big Sales Prediction project aimed to accurately forecast sales figures by leveraging the RFR because of its robustness in handling complex datasets
•The Random Forest Regressor from sk-learn was then applied to predict sales, with model performance rigorously evaluated using the MAE metric.

Indian Oil Corporation Ltd. Lube Blending Plant, Chennai | Mr. Nibil PS, Lube Safety Manager. [Dec'23 - Jan'24]
Worked as a student intern at IOCL, Chennai | Learned about Pump House functioning, small can filling, barrel filling, LOBS, QC Lab

PROJECTS

Understanding the spatiotemporal variation of LAI and its correlation with climatic variables in India & neighboring countries in past 22 years
• Investigated the role of Leaf Area Index (LAI) as a key indicator for assessing vegetation-climate interactions across five agroclimatic regions of India.
• Analyzed 22 years of satellite data to examine temporal trends in LAI and its relationship with hydrometeorological variables: Prec, Temp, PDSI
• Performed correlation and spatial analysis to identify dominant climatic drivers influencing vegetation dynamics in each agroclimatic zone.
• Identified instances where temp positively influenced vegetation growth, while precipitation occasionally exhibited negative impact on vegetation dynamics

Bias Correction of Monthly Precipitation Estimates using Deep Learning | CNN
•Preprocessing involved standardizing grid values to a 32×32 format, resulting in generation of 1,290 input samples for each dataset spanning over 30 years.
•Then data was strategically split, with 80% allocated for training and the remaining 20% reserved for testing the deep learning model.
•The model demonstrated strong performance, as Pearson's Correlation Coefficient yielded a correlation value of 0.87 for model data and 0.73 for bias data.

Safe Building Analysis using Google Street View Images Classification by implementing YOLOv8n and ResNet50
•Fine-tuned a YOLOv8n-cls pretrained on ImageNet to save training time and improve performance with a small dataset gives 77.6% accuracy.
•Then data was strategically split, with 80% allocated for training and the remaining 20% reserved for testing the deep learning model.
•ResNet50, grayscale images converted to 3-channel format (to match ResNet50's input format) then it is resized to 244*244*3 pixels attains 67.8% accuracy.

PUBLICATION

Singh, Y., et al., 2025. Understanding the spatiotemporal variation of leaf area index and its correlation with climate variables in India and neighboring countries over the past two decades. *Advances in Space Science* (Reviewing Stage).

CONFERENCE

Student Delegate | Tropmet 2024 | Indian Meteorological Society and NIT Rourkela [Dec 2024]
Presentation on 'Understanding the Climatic impacts on Vegetation using LAI (Leaf Area Index) as a key variable'.

SKILLS

Technical Skills: Remote Sensing & GIS, Satellite Data Analysis, Time Series Analysis, Spatial Data Analysis, Climate Data Processing, Climate risk, Carbon Accounting | **Analytical Skills:** Correlation & Trend Analysis | **Language and DBMS:** Python.
Core Skills - Net zero Analysis, ESG, TCFD & BRSR reports, Power BI, Excel, PowerPoint, Word, QGIS, Earth Engine, Java Script API.

COURSEWORK INFORMATION

Satellite Remote Sensing | Climate Risk Adaptation Mitigation and Sustainable Development | Carbon Cycle and Global Climate Change | Data Analytics in Earth System Science | Physical Oceanography and Climate Change | Computational Methods for Earth System Science | Tectonics and Fluvial Dynamics | Design Lab | Global Climate System and Cloud Precipitation Processes

CERTIFICATIONS

Python for Everybody | Fundamentals of python programming authorized by University of Michigan.
Data Analysis with Python | Importing Dataset | Data Wrangling | Exploratory Data Analysis | Model Development | Model Evaluation and Refinement

POSITION OF RESPONSIBILITY

Student Coordinator | Founding Member | Training and Placement Cell, CORAL | IIT Kharagpur [Nov 2023 – May 2025]
•Worked closely with esteemed professors, (CDC Coordinators, CORAL), engaging with potential employers from diverse backgrounds, fostering connections between Earth System Science domain and its growing industry with a focus on AI, ML and Generative AI.
•Represented the CORAL at Academia-Industry Conclave (Dec '23, July '24), organized by Career Development Centre, IIT Kharagpur.